Infrastruktur Pasif web app structure (under development)



Excel Data Construct

No	Field	Satuan	Data Type	Limitation/Validation Requirement	Nature of field
1	Kode/ID menara	-	String	50 Character limit	Mandatory
2	Nama pemilik menara telekomunikasi	-	String	50 Character limit	Mandatory
3	Tahun Pembangunan Menara Telekomunikasi	-	Positive Integer	Four-digit integer, from 1900 through the current year, inclusive	Mandatory
4	Penyelenggara Jaringan Telekomunikasi yang menyewa menara telekomunikasi	-	String	50 character limit. Each row represents one tenant relationship	Mandatory
5	Longitude	Degree	Float	6 numbers behind the dot. Can only be filled between 95 to 141	Mandatory
6	Latitude	Degree	Float	6 numbers behind the dot. Can only be filled between -11 to 6	Mandatory
7	Alamat	-	String	50 Character limit	Mandatory
8	Jenis Menara Telekomunikasi	?	String	50 Character limit	Mandatory
9	Tinggi Menara Telekomunikasi (AGL)	meter	Positive Integer	3 Character limit	Mandatory
10	Kapasitas maksimal menara telekomunikasi	?	String	50 Character limit	Mandatory
11	Kapasitas menara telekomunikasi yang disewakan	?	String	50 Character limit	Mandatory
12	Kapasitas menara telekomunikasi yang digunakan sendiri	?	String	50 Character limit	Mandatory

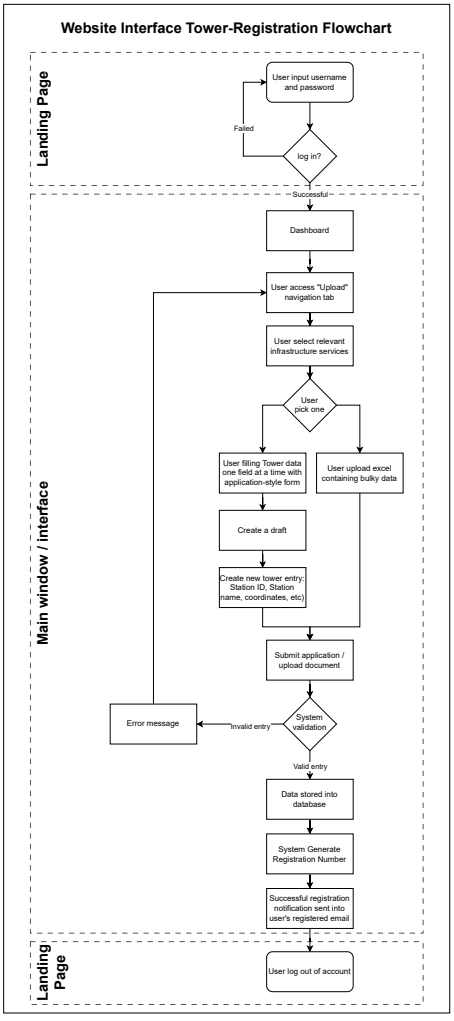
Account Role

Functionality	Normal User	Web Manager	Super Admin
Upload excel files	Yes	Yes	Yes
View own submission history and details	Yes	Yes	Yes
View all users' submission history and details	No	Yes	Yes
View tower records	Own records	All records	All records
Download tower data	Own records	All records	All records
Delete individual tower records (soft delete)	Own records	All records	All records
Delete tower data in bulk	Own records	All records	All records
View audit logs	No	Yes	Yes
Change own password	Yes	Yes	Yes
Create user accounts	No	No	Yes
Assign or change account types	No	No	Yes
Change other user's passwords	No	No	Yes

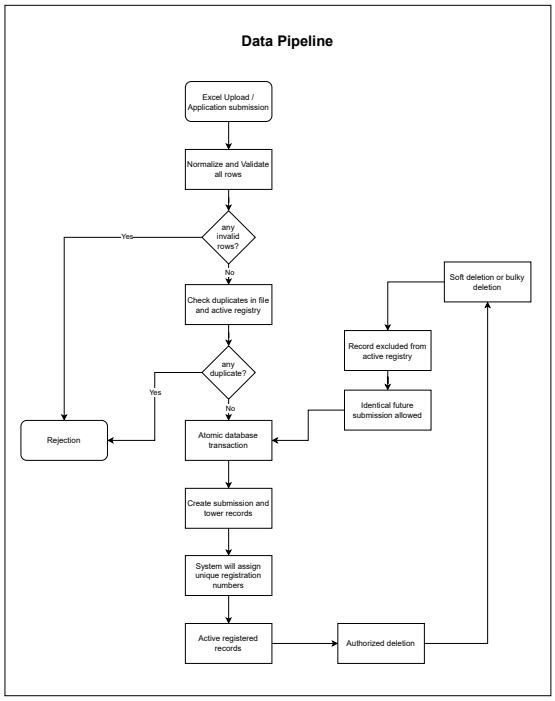
Account Managements

Feature	Web Manager	Super Admin
View all submission history and details	Yes	Yes
Create user accounts	No	Yes
Assign the `User` role to new accounts	No	Yes
Assign the `Web Manager` role	No	Yes
Assign the `Super Admin` role	No	Yes
Change existing users' roles	No	Yes
Reset other users' passwords	No	Yes
Manage own password	Yes	Yes
Deactivate accounts	No	Yes
Reactivate accounts	No	Yes

Website Interface Tower-Registration Flowchart



Data Pipeline



UI Component

Landing page	1. Icon / logo display 2. Groupbox container 3. Label 4. Text field / line edit area
Main window / interface	• Web Header containing icon/logo • Retractable navigation tab (left side) containing: Dashboard, Data Submission, Account, and Log out button. • App version and Copyright claim at bottom side of the web Inside Dashboard tab: 1. Map displaying each tower data 2. A lot of query data fields 3. Configurable table display with pagination setting and download pushbutton 4. Export pushbutton Inside Data Submission tab: 1. Dropdown / radio button for application selection (upload-excel or form-based-application) 2. for upload excel selection: o label o text field / line edit o pushbutton 3. for form-based-application selection: o label o text field / line edit o pushbutton o radio button o date picker Inside Account tab: 1. label 2. text fields 3. pushbutton